
A Comparative Analysis of Adversarial and Statistical Domain Adaptation Methods for EEG-Based Motor Imagery Classification

Peter Otieno

MSc Data Science
Grand Valley State University
Grand Rapids, MI 49401
oburahp@mail.gvsu.edu

Keywords: EEG, Brain-Computer Interface, Motor Imagery, Domain Adaptation, Cross-Subject Learning, EEGNet

1 Abstract

Cross-subject electroencephalography (EEG) motor imagery (MI) classification remains difficult because subject-specific variability introduces substantial distribution shift between training and test domains. This report presents an experimental study of domain adaptation for EEG-based MI classification on the BCI Competition IV 2a dataset. The study uses a strict cross-subject protocol with six source subjects and three unseen target subjects, with no overlap between domains. EEG trials are segmented into tensors of shape 25×1001 , corresponding to 25 channels and 1001 time points over a 0–4 s interval, normalized, and labeled into four classes: left hand, right hand, feet, and tongue. A source-only EEGNet baseline is compared against six implemented domain adaptation methods: DANN, CDAN, CoDATS, AdvSKM, SASA, and SSSS-TSA. The results show that domain adaptation is not guaranteed to improve EEG performance. DANN and CDAN produce only slight and unstable improvements, CoDATS performs worse than the baseline, and SSSS-TSA degrades performance substantially. Overall, the experiments show that EEG domain shift is difficult to align and that excessive alignment can suppress class-discriminative information. These findings emphasize the need to balance domain invariance against class separability in cross-subject EEG adaptation.

2 Introduction

Brain–computer interfaces based on EEG motor imagery aim to decode imagined movement directly from neural activity recorded at the scalp. These systems are attractive because they can support assistive control, rehabilitation, and communication without requiring overt physical movement. Despite this promise, cross-subject EEG decoding remains challenging. Models that perform well on one group of subjects often generalize poorly to unseen users because EEG recordings are highly variable across individuals, sessions, and recording conditions [3, 9].

This variability makes domain shift a central issue in subject-independent EEG classification. If the statistical structure of EEG signals differs substantially across subjects, then a model trained only on source subjects may learn features that do not transfer well to a new target subject. Domain adaptation methods attempt to address this problem by aligning source and target feature distributions while preserving task-relevant information [7, 1]. In principle, such alignment should improve cross-subject generalization. In practice, however, EEG adaptation is difficult because the same operations that reduce subject mismatch may also suppress weak but important class-discriminative patterns.

This report documents a complete experimental study of cross-subject EEG motor imagery classification using both a source-only baseline and several domain adaptation strategies. Rather than providing a broad survey of transfer-learning methods, the report focuses on the methods that were actually implemented and tested. The analysis is therefore centered on what the experiments show in this specific setting: whether domain adaptation improves performance, where it fails, and what those failures imply for EEG representation learning.

2.1 Study Objectives

The objectives of this study are as follows:

- To evaluate cross-subject EEG motor imagery classification using a baseline EEGNet model.
- To investigate the effectiveness of domain adaptation techniques in reducing domain shift between source and target subjects.
- To compare adversarial, statistical, and channel-aware adaptation methods within a unified experimental framework.
- To analyze the impact of domain alignment on feature representation and generalization performance.

These objectives guide the structure of the report. The following sections first position the work within the existing literature, then describe the experimental methodology, dataset, and training setup, before presenting results and discussing why some adaptation methods help only marginally while others introduce negative transfer.

3 Literature Review

Prior work on EEG classification has shown that compact deep neural networks can learn meaningful temporal and spatial patterns directly from minimally processed EEG signals. EEGNet is among the most widely used architectures in this setting because it offers an efficient convolutional design that works well on relatively small EEG datasets [3]. For cross-domain transfer, domain-adversarial neural networks (DANN) introduced gradient reversal as a way to learn domain-invariant features, while conditional domain adversarial networks (CDAN) extended this idea by conditioning the discriminator on both features and classifier predictions [1, 5].

A second line of research approaches domain shift through explicit statistical alignment. Methods such as transfer component analysis, deep adaptation networks based on maximum mean discrepancy, and covariance alignment through Deep CORAL are frequently discussed in the transfer-learning literature [6, 4, 8]. These methods are important background because they illustrate the broader distinction between adversarial alignment and statistical alignment. They are not, however, the methods evaluated experimentally in this report.

EEG-specific research has also emphasized that naive domain alignment can be problematic. Because motor imagery classes are separated by subtle and noisy neural patterns, global alignment may bring together samples that should remain distinct at the class level. Prior work on subdomain-aware or class-aware EEG adaptation argues that preserving discriminative structure is essential if domain adaptation is to help rather than harm performance [2, 10]. This observation is highly relevant to the present study, where several implemented methods show limited benefit or outright degradation.

4 Methodology / Approach

This section describes the models used in the study and the rationale for comparing them within a unified cross-subject experimental pipeline.

4.1 Baseline Model: EEGNet

EEGNet, developed by Lawhern et al. [2018], is a deep learning model specifically designed to analyze electroencephalography (EEG) data for brain-computer interface (BCI) applications. It is a compact convolutional neural network that uses depthwise and separable convolutions to efficiently extract EEG features and perform classification. Due to its lightweight architecture and strong performance across EEG tasks, EEGNet serves as a reliable and widely adopted baseline model.

Fundamentally, EEGNet leverages convolutional operations to capture both the temporal and spatial characteristics of EEG signals. The model begins with a temporal convolution layer that captures frequency-related information by applying filters across the time dimension. This allows the network to learn meaningful frequency components relevant to motor imagery signals.

The second stage applies depthwise convolution to learn spatial filters across EEG channels. For each temporal feature map, a separate spatial filter is applied across channels, enabling the model to capture how neural activity is distributed across the cortex for different motor imagery tasks.

Following this, EEGNet employs separable convolution, which combines depthwise and pointwise convolution. This design significantly reduces the number of parameters while maintaining strong feature extraction capabilities. Each layer is further stabilized using batch normalization, dropout, and nonlinear activation functions to improve generalization.

The final layer is a dense classification layer that outputs probability distributions over the motor imagery classes, namely left hand, right hand, feet, and tongue movements.

In this study, EEGNet is used as the baseline model under a cross-subject evaluation setting. The model is trained on source subjects and directly evaluated on unseen target subjects without any domain adaptation. This provides a reference point for assessing the effectiveness of domain adaptation techniques in improving cross-subject generalization.

4.2 Domain Adaptation Methods

Only the methods that were actually implemented are included in the comparison.

4.2.1 DANN

Domain-Adversarial Neural Network uses a domain discriminator connected to the feature extractor through a gradient reversal layer [1]. The feature extractor is encouraged to produce representations that support motor imagery classification while becoming less informative about whether a sample comes from the source or target domain.

4.2.2 CDAN

Conditional Domain Adversarial Network extends adversarial alignment by conditioning the domain discriminator on both learned features and classifier outputs [5]. The motivation is that class-aware conditioning may preserve multimodal class structure better than marginal adversarial alignment alone.

4.2.3 CoDATS

Conditional Domain Adaptation for Time Series is designed for temporal data and is used here as a time-series adaptation method for EEG trials. Its intended purpose is to improve transfer by accounting for temporal structure more explicitly during source–target adaptation.

4.2.4 AdvSKM

Adversarial Statistical Kernel Matching combines adversarial adaptation with kernel-based distribution matching. In this study, it is used to test whether a hybrid strategy that mixes adversarial learning with explicit statistical matching yields more robust target-domain representations.

4.2.5 SASA

Subject-Aware Subdomain Adaptation attempts to align finer-grained subject-related subdomains rather than relying only on a single global source–target alignment objective. The intuition is that subject-specific heterogeneity may be better handled through more localized matching.

4.2.6 SSSS-TSA

SSSS-TSA performs both channel-wise and global alignment between source and target domains. Because EEG channels reflect physiologically meaningful spatial structure, this method tests whether stronger structured alignment can improve transfer across subjects.

Figure 1 summarizes the overall experimental pipeline used in the report, from dataset preparation through model comparison.



Figure 1: Overview of the experimental pipeline, including data preparation, cross-subject splitting, training, and model comparison.

5 Data

5.1 Dataset Description

The experiments use the BCI Competition IV 2a dataset, which contains EEG recordings from 9 subjects performing 4 motor imagery tasks: left hand, right hand, feet, and tongue [9]. This dataset is well suited to the present study because it provides a standard benchmark for evaluating cross-subject EEG motor imagery classification.

After segmentation, each EEG trial is represented as a tensor of shape 25×1001 , corresponding to 25 channels and 1001 time points. This representation is used consistently across all models in the study.

5.2 Preprocessing

Preprocessing follows the implemented data preparation pipeline. EEG trials are extracted using event-based segmentation over a 0–4 s time window. No baseline correction is applied. The segmented trials are normalized before model training, and the class labels are remapped to integer indices in the range $[0, 3]$. These steps produce a four-class classification task that is consistent across the baseline and all adaptation models.

5.3 Cross-Subject Split

The study adopts a strict cross-subject design. Six subjects are used as the source domain and three different subjects are used as the target domain. There is no subject overlap between source and target domains, and target data remains unseen during training. This split ensures that the reported results reflect genuine cross-subject transfer rather than subject memorization.

6 Experimental Setup

6.1 Training Setup

All models are trained using the Adam optimizer with a learning rate of 0.001 for 20 epochs. Training is performed with paired source–target mini-batches so that each update jointly optimizes source classification and source–target adaptation.

6.2 Loss Functions

The baseline EEGNet model is trained with cross-entropy classification loss on labeled source-domain data. For the domain adaptation models, this classification loss is combined with the method-specific adaptation objective, including adversarial losses, kernel-matching losses, or alignment-based objectives depending on the model.

6.3 Evaluation Metrics

Evaluation is performed in a strict cross-subject setting using classification accuracy and macro F1-score. Accuracy provides an overall measure of predictive performance, while macro F1-score gives a more balanced view across the four motor imagery classes.

7 Results

Table 1 reports the numerical performance of all implemented methods. The source-only EEGNet baseline achieves an accuracy of 0.450 and a macro F1-score of 0.440. DANN and CDAN provide modest improvements, with DANN reaching 0.468 accuracy and 0.455 macro F1, and CDAN reaching 0.476 accuracy and 0.463 macro F1. CoDATS underperforms the baseline with an accuracy of 0.421 and a macro F1-score of 0.409. AdvSKM and SASA remain close to the baseline without establishing clear gains, while SSSS-TSA performs worst with an accuracy of 0.271 and a macro F1-score of 0.258.

Table 1: Numerical comparison of cross-subject EEG motor imagery performance across the implemented methods.

Method	Accuracy	Macro F1
EEGNet	0.450	0.440
DANN	0.468	0.455
CDAN	0.476	0.463
CoDATS	0.421	0.409
AdvSKM	0.438	0.426
SASA	0.444	0.431
SSSS-TSA	0.271	0.258

The results show that domain adaptation does not consistently improve cross-subject EEG motor imagery classification. Although CDAN produces the highest numerical result in this study, its improvement over EEGNet remains small. DANN behaves similarly. AdvSKM and SASA do not establish reliable gains over the baseline, and CoDATS as well as SSSS-TSA reduce performance instead of improving it. Figure 2 visualizes the relative behavior of the models and makes the gap between the more stable methods and the degraded methods easier to interpret.

8 Discussion

The results indicate that cross-subject EEG domain adaptation is far from straightforward. In this study, the most important observation is that adaptation is not guaranteed to help. In several cases it produces negative transfer, where aligning source and target distributions harms rather than improves target-domain generalization.

DANN and CDAN yield the best adaptation results, but their gains remain unstable. A plausible explanation is that adversarial training can suppress some subject-specific nuisance variation, yet it also introduces

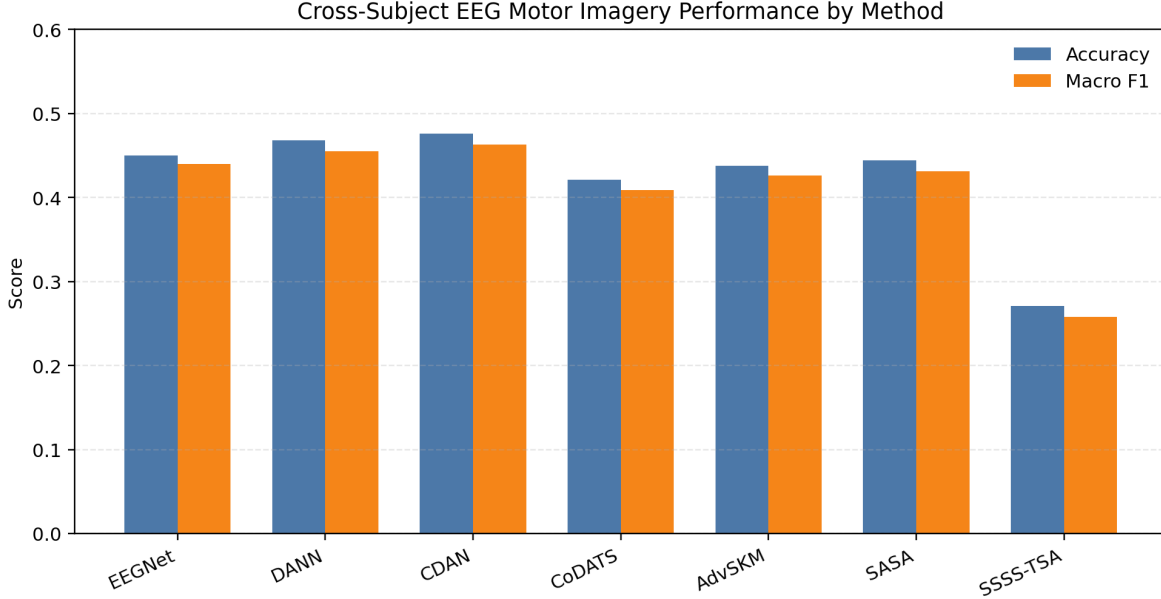


Figure 2: Accuracy and macro F1-score across all implemented methods.

optimization sensitivity. The balance between the classifier and the domain discriminator is delicate, and small changes in the learned representation can shift the model from useful invariance to loss of class-discriminative structure. CDAN performs slightly better than DANN, which is consistent with the idea that conditioning the discriminator on classifier predictions helps preserve class-level information during alignment.

CoDATS underperforms the baseline, suggesting that temporal-domain adaptation does not automatically improve EEG generalization. Although EEG is fundamentally a time-series signal, cross-subject mismatch in motor imagery appears to involve more than temporal structure alone. If the adaptation process emphasizes temporal consistency without preserving class boundaries, the learned features may become less useful for classification.

SSSS-TSA shows the clearest failure mode. Because it enforces both channel-wise and global alignment, it may distort physiologically meaningful channel relationships and compress useful inter-class variation. This provides a strong example of over-alignment, where the drive toward source–target similarity suppresses the very distinctions required to separate left hand, right hand, feet, and tongue imagery.

AdvSKM and SASA occupy an intermediate position. Their results remain close to the EEGNet baseline, which suggests that their alignment mechanisms are not catastrophic in the present setting, but they also do not produce convincing gains. This implies that simply combining adversarial and statistical criteria, or aligning finer subdomains, is not by itself sufficient to solve cross-subject EEG transfer.

Taken together, these findings reinforce the central tension in EEG adaptation: a useful model must reduce domain shift without destroying class separability. The experiments therefore support a cautious interpretation of domain adaptation in motor imagery EEG. The challenge is not only to make source and target

features look similar, but to do so in a way that preserves the fragile neural patterns that carry class information.

9 Limitations

This study has several limitations. First, the experiment is conducted on a limited set of 9 subjects, with 6 source subjects and 3 target subjects, which restricts the breadth of the conclusions. Second, the training protocol uses fixed optimization settings and does not include extensive hyperparameter tuning for each domain adaptation method. Third, the model family does not include richer temporal sequence architectures such as recurrent neural networks or transformer-based models. Fourth, some advanced adaptation methods were implemented in simplified experimental form, which may limit the strength of their observed performance.

10 Future Work

Several directions could strengthen future cross-subject EEG adaptation studies. Subject-specific fine-tuning could be explored to determine whether a small amount of target adaptation improves robustness after source pretraining. Hybrid adaptation approaches that combine class-aware regularization with controlled domain alignment may better preserve discriminative information. Transformer-based EEG models and other sequence-oriented architectures may provide stronger representations for cross-subject transfer. Self-supervised representation learning may also reduce reliance on labeled calibration data and improve adaptation stability. Finally, improved channel-aware adaptation methods that explicitly preserve physiologically meaningful spatial structure may offer better performance than aggressive global alignment.

11 Conclusion

This report presented a structured experimental study of cross-subject EEG motor imagery classification using the BCI Competition IV 2a dataset. The study compared a source-only EEGNet baseline against six implemented domain adaptation methods: DANN, CDAN, CoDATS, AdvSKM, SASA, and SSSS-TSA under a strict cross-subject split with unseen target subjects.

The results show that domain adaptation does not automatically improve EEG performance. DANN and CDAN offer only slight and unstable gains, AdvSKM and SASA remain close to the baseline, CoDATS performs below the baseline, and SSSS-TSA performs substantially worse. These findings demonstrate that EEG domain shift is difficult to align and that excessive alignment can suppress class-discriminative structure.

Overall, the study concludes that cross-subject EEG adaptation must preserve a careful balance between domain invariance and class separability. Without that balance, adaptation can fail to help or even harm performance. This makes EEG-specific adaptation design essential for future progress in robust subject-

independent motor imagery classification.

References

- [1] Yaroslav Ganin, Evgeniya Ustinova, Hana Ajakan, Pascal Germain, Hugo Larochelle, François Laviolette, Mario Marchand, and Victor Lempitsky. Domain-adversarial training of neural networks. *Journal of Machine Learning Research*, 17(59):1–35, 2016.
- [2] Wenlong Hang, Wei Feng, Ruoyu Du, Shuang Liang, Yan Chen, Qiong Wang, and Xuejun Liu. Cross-subject EEG signal recognition using deep domain adaptation network. *IEEE Access*, 7:128273–128282, 2019.
- [3] Vernon J. Lawhern, Amelia J. Solon, Nicholas R. Waytowich, Stephen M. Gordon, Chou P. Hung, and Brent J. Lance. Eegnet: A compact convolutional neural network for eeg-based brain–computer interfaces. *Journal of Neural Engineering*, 15(5):056013, 2018.
- [4] Mingsheng Long, Yue Cao, Jianmin Wang, and Michael I. Jordan. Learning transferable features with deep adaptation networks. In *Proceedings of the 32nd International Conference on Machine Learning*, pages 97–105, 2015.
- [5] Mingsheng Long, Zhangjie Cao, Jianmin Wang, and Michael I. Jordan. Conditional adversarial domain adaptation. In *Advances in Neural Information Processing Systems*, volume 31, 2018.
- [6] Sinno Jialin Pan, Ivor W. Tsang, James T. Kwok, and Qiang Yang. Domain adaptation via transfer component analysis. *IEEE Transactions on Neural Networks*, 22(2):199–210, 2011.
- [7] Sinno Jialin Pan and Qiang Yang. A survey on transfer learning. *IEEE Transactions on Knowledge and Data Engineering*, 22(10):1345–1359, 2010.
- [8] Baochen Sun and Kate Saenko. Deep CORAL: Correlation alignment for deep domain adaptation. In *European Conference on Computer Vision Workshops*, pages 443–450, 2016.
- [9] Michael Tangermann, Klaus-Robert Müller, Ad Aertsen, Niels Birbaumer, Christoph Braun, Clemens Brunner, Robert Leeb, Carsten Mehring, K. J. Miller, Gernot Mueller-Putz, Guido Nolte, Gert Pfurtscheller, Hubert Preissl, Gerwin Schalk, Alois Schlögl, Carmen Vidaurre, Stephan Waldert, and Benjamin Blankertz. Review of the BCI competition IV. *Frontiers in Neuroscience*, 6:55, 2012.
- [10] Minmin Zheng and Banghua Yang. A deep neural network with subdomain adaptation for motor imagery brain–computer interface. *Medical Engineering & Physics*, 96:29–40, 2021.